

## Types of Dependencies in DBMS

A **dependency** in DBMS means that the value of one attribute depends on the value of another attribute.

### 1. Functional Dependency (FD)

When one attribute uniquely determines another attribute.

**Notation:**  $A \rightarrow B$

**Example:**

$\text{Student\_ID} \rightarrow \text{Student\_Name}$

If we know the Student ID, we can determine the Student Name.

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### 2. Partial Dependency

When a non-key attribute depends on **only a part of a composite primary key**, it is called Partial Dependency.

**Example:**

Composite key = (Student\_ID, Course\_ID)

If:

$\text{Student\_ID} \rightarrow \text{Student\_Name}$

then Student\_Name depends only on part of the composite key.

**Removed in: 2NF**

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### 3. Transitive Dependency

When a non-key attribute depends on another non-key attribute.

**Example:**

$\text{Student\_ID} \rightarrow \text{Department\_ID}$

$\text{Department\_ID} \rightarrow \text{Department\_Name}$

Therefore:

$\text{Student\_ID} \rightarrow \text{Department\_Name}$

Removed in: 3NF

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#### 4. Multi-Valued Dependency (MVD)

When one attribute determines **multiple independent values** of another attribute.

**Notation:**  $A \twoheadrightarrow B$

**Example:**

Student  $\twoheadrightarrow$  Hobby

A student can have multiple hobbies.

**Handled in: 4NF**

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#### 5. Join Dependency

When a table can be divided into multiple smaller tables and the original table can be reconstructed by joining them.

**Handled in: 5NF**

#### Quick Revision

| Dependency              | Meaning                                       | Normal Form |
|-------------------------|-----------------------------------------------|-------------|
| Functional Dependency   | One attribute determines another              | 1NF–3NF     |
| Partial Dependency      | Depends on part of a composite key            | 2NF         |
| Transitive Dependency   | Non-key depends on non-key                    | 3NF         |
| Multi-Valued Dependency | One attribute has multiple independent values | 4NF         |
| Join Dependency         | Original relation reconstructed using joins   | 5NF         |

**Easy way to remember:**

**2NF  $\rightarrow$  Partial**

**3NF  $\rightarrow$  Transitive**

**4NF  $\rightarrow$  Multi-Valued**

**5NF  $\rightarrow$  Join**